Math 74: Algebraic Topology

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> **Problem 1**. Compute the simplicial homology for $\mathbb{RP}^2$ using a Δ-complex structure on it.

Solution.

$\mathbb{RP}^2$ as a Δ-complex is formed by two 0-simplices, three 1-simplices and two 2-simplices.

$$\begin{tikzcd}
v_0 \arrow[dr, phantom, "U", near start] & v_1 \arrow[l, "b"'] \arrow[d, "a"] \\
v_1 \arrow[u, "a"] \arrow[ur, "c"] \arrow[r, "b"'] \arrow[ur, phantom, "L"', near end] & v_0
\end{tikzcd}$$

We have,

- $\Delta_0(\mathbb{RP}^2) = \mathbb{Z}\langle v_0 \rangle \oplus \mathbb{Z}\langle v_1 \rangle$
- $\Delta_1(\mathbb{RP}^2) = \mathbb{Z}\langle a \rangle \oplus \mathbb{Z}\langle b \rangle \oplus \mathbb{Z}\langle c \rangle$
- $\Delta_2(\mathbb{RP}^2) = \mathbb{Z}\langle U \rangle \oplus \mathbb{Z}\langle L \rangle$

$$0 \xrightarrow{\partial_3} \Delta_2(\mathbb{RP}^2) \xrightarrow{\partial_2} \Delta_1(\mathbb{RP}^2) \xrightarrow{\partial_1} \Delta_0(\mathbb{RP}^2) \xrightarrow{\partial_0} 0$$

The homologies are,

- $H_0^{\Delta}(\mathbb{RP}^2) = \frac{\ker(\partial_0)}{\operatorname{im}(\partial_1)} = \frac{\mathbb{Z}\langle v_0 \rangle \oplus \mathbb{Z}\langle v_1 \rangle}{\langle v_0 - v_1 \rangle} \cong \mathbb{Z}\langle v_0 \rangle \cong \mathbb{Z}$
- $H_1^{\Delta}(\mathbb{RP}^2) = \frac{\ker(\partial_1)}{\operatorname{im}(\partial_2)} = \frac{\mathbb{Z}\langle a-b \rangle \oplus \mathbb{Z}\langle c \rangle}{\mathbb{Z}\langle a-b-c, a-b+c \rangle}$

  Let $x = a - b$, $y = c$. Then we have,

  $$x + y = 0, \quad x - y = 0 \implies 2x = 2y = 0$$

  i.e. both x and y have order 2. And,

  $$x - y = 0 \implies x = y$$

So,

$$H_1^\Delta(\mathbb{RP}^2) = \frac{\mathbb{Z}\langle a-b\rangle \oplus \mathbb{Z}\langle c\rangle}{\mathbb{Z}\langle a-b-c,\, a-b+c\rangle} \cong \frac{\mathbb{Z}\langle x\rangle}{2x} \cong \mathbb{Z}/2\mathbb{Z}$$

- $H_2^\Delta(\mathbb{RP}^2) = \frac{\ker(\partial_2)}{\mathrm{im}(\partial_3)} = \frac{\{0\}}{\{0\}} \cong \{0\}$
- $H_n^\Delta = 0$ if $n \geq 3$.

Problem 2. Prove directly that singular homology is an invariant of homeomorphism class of the space. Here 'directly' means without referring to the fact that singular homology is an invariant of homotopy equivalence class of the space.

Solution.

Let $f : X \to Y$ be a homeomorphism. Then for each singular n-simplex $\sigma^n : \Delta^n \to X$ we define $f'_n(\sigma^n) = f \circ \sigma^n$, i.e. $f'(\sigma^n) : \Delta^n \to Y$. We will show that the following diagram commutes,

$$\begin{array}{ccc} C_n(X) & \xrightarrow{\partial_n} & C_{n-1}(X) \\ \big\downarrow{\scriptstyle f'_n} & & \big\downarrow{\scriptstyle f'_{n-1}} \\ C_n(Y) & \xrightarrow{\partial_n} & C_{n-1}(Y) \end{array}$$

So,

$$\begin{aligned} f'_{n-1} \circ \partial_n(\sigma^n) &= f'_{n-1}\left(\sum_i (-1)^i \sigma^n \mid_{[v_0,\ldots,\widehat{v_i},\ldots,v_n]}\right) = \sum_i (-1)^i f \circ \sigma^n \mid_{[v_0,\ldots,\widehat{v_i},\ldots,v_n]} \\ &= \sum_i (-1)^i (f \circ \sigma^n) \mid_{[v_0,\ldots,\widehat{v_i},\ldots,v_n]} \\ &= \partial_n(f \circ \sigma^n) = \partial_n \circ f'_n(\sigma^n) \end{aligned}$$

Note that since $X \cong Y$ the map $f^{-1} : Y \to X$ induces the inverse chain maps

$$\begin{array}{ccc} C_n(Y) & \xrightarrow{\partial_n} & C_{n-1}(Y) \\ \big\downarrow{\scriptstyle (f^{-1})'_n} & & \big\downarrow{\scriptstyle (f^{-1})'_{n-1}} \\ C_n(X) & \xrightarrow{\partial_n} & C_{n-1}(X) \end{array}$$

Thus making the chain complexes isomorphic. Now show that the induced maps $\widetilde{f} : H_n(X) \to H_n(Y)$, and $\widetilde{f^{-1}} : H_n(Y) \to H_n(X)$ on the singular homology groups are isomorphisms.

- **Well-defined:** Assume $[\sigma^n_\alpha] = [\sigma^n_\beta] \in H_n(X)$. Then, $\sigma^n_\alpha - \sigma^n_\beta \in \operatorname{im}(\partial_{n+1})$, that is there exists $\tau \in C_{n+1}(X)$ such that

  $$\sigma^n_\alpha - \sigma^n_\beta = \partial_{n+1}(\tau) \implies f'_n(\sigma^n_\alpha) - f'_n(\sigma^n_\beta) = f'_n(\partial_{n+1}(\tau)) = \partial_{n+1}(f'_{n+1}(\tau))$$

  Hence, $[f'_n(\sigma^n_\alpha)] = [f'_n(\sigma^n_\beta)] \in H_n(Y)$

- **Isomorphism:** For $\sigma^n \in C_n(X)$ and $\tau^n \in C_n(Y)$, we have

$$\widetilde{f^{-1}} \circ \widetilde{f}([\sigma^n]) = [(f^{-1})'_n \circ f'_n(\sigma^n)] = [\sigma^n] = \mathrm{id}_{H_n(X)}([\sigma^n])$$

  and

$$\widetilde{f} \circ \widetilde{f^{-1}}([\tau^n]) = [f'_n \circ (f^{-1})'_n([\tau^n])] = [\tau^n] = \mathrm{id}_{H_n(Y)}([\tau^n])$$

Hence $\widetilde{f}$ is an isomorphism and the singular homology is an invariant of the homeomorphism class of the topological space.

Problem 3. Compute the singular homology groups of a point.

Solution.

Let x_0 be a point. Then the constant map $\sigma_n : \Delta^n \to X$ gives us a n-simplex for all n since constant maps are continuous. Since the restriction of the constant map is the constant map we get,

$$\partial_n(\sigma_n) = \sum_{i=0}^{n} (-1)^i \sigma_{n-1} = \sigma_{n-1} \sum_{i=0}^{n} (-1)^i = \begin{cases} 0, & \text{n odd} \\ \sigma_{n-1}, & \text{n even} \end{cases}$$

Thus we get,

$$\cdots \xrightarrow{\partial_3} C_2(x_0) \xrightarrow{\partial_2} C_1(x_0) \xrightarrow{\partial_1} C_0(x_0) \xrightarrow{\partial_0} 0$$
$$\cdots \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$$

Then for odd n we have, $\ker(\partial_n) = \mathbb{Z}$ and $\operatorname{im}(\partial_{n+1}) = \operatorname{im}(\operatorname{id}_{C_{n-1}}) = \mathbb{Z}$, so

$$H_n(x_0) = \frac{\ker(\partial_n)}{\operatorname{im}(\partial_{n+1})} = \frac{\mathbb{Z}}{\mathbb{Z}} = \{0\}$$

For even $n > 0$, we have $\ker(\partial_n) = 0$, so

$$H_n(x_0) = \frac{\ker(\partial_n)}{\operatorname{im}(\partial_{n+1})} = \{0\}$$

For $n = 0$, we have,

$$H_0(x_0) = \frac{\ker(\partial_0)}{\operatorname{im}(\partial_1)} = \frac{\mathbb{Z}}{0} = \mathbb{Z}$$

Hence, the singular homology groups of a point are 0 for $n > 0$ and $\mathbb{Z}$ for $n = 0$.

Problem 4. Compute the homology groups of the Δ-complex X obtained from Δ^n by identifying all faces of the same dimension. Thus X has a single k simplex for each $k \leq n$.

Solution.

We get

$$\cdots \xrightarrow{\partial_3} \Delta^2(X) \xrightarrow{\partial_2} \Delta^1(X) \xrightarrow{\partial_1} \Delta^0(X) \xrightarrow{\partial_0} 0$$

where for $k \leq n$,

$$\partial_k(\sigma_k) = \sum_i (-1)^i \sigma_{k-1} = \sigma_{k-1} \sum_i (-1)^i = \begin{cases} 0, & \text{k odd} \\ \sigma_{k-1}, & \text{k even} \end{cases}$$

and for $k > n$ $\partial_k(\sigma_k) = 0$. Then the homology groups are as follows,

- $H_0(X) = \ker(\partial_0)/\operatorname{im}(\partial_1) = \mathbb{Z}/0 \cong \mathbb{Z}$
- If k odd, $k < n$
$$H_k(X) = \ker(\partial_k)/\operatorname{im}(\partial_{k+1}) = \mathbb{Z}/\mathbb{Z} \cong 0$$
- If $k = n$, n odd
$$H_n(X) = \ker(\partial_n)/\operatorname{im}(\partial_{n+1}) = \mathbb{Z}/0 \cong \mathbb{Z}$$
- If k even, $k \leq n$
$$H_k(X) = \ker(\partial_k)/\operatorname{im}(\partial_{k+1}) = 0/0 \cong 0$$
- If $k > n$
$$H_k(X) = \ker(\partial_k)/\operatorname{im}(\partial_{k+1}) = 0/0 \cong 0$$

Problem 5. Compute $H_i(D^n, S^{n-1})$ for all i and n.

Solution.

We have the long exact sequence,

$$\cdots \to H_i(S^{n-1}) \xrightarrow{i_\star} H_i(D^n) \xrightarrow{\pi_\star} H_i(D^n, S^{n-1}) \xrightarrow{\partial_i} H_{i-1}(S^{n-1}) \xrightarrow{i_\star} H_{i-1}(D^n) \to \cdots$$

Since D^n is contractible and homology is an invariant of the homotopy equivalence class of the space, we have $H_i(D^n) = 0$ for $i > 0$ and $H_0(D^n) = \mathbb{Z}$. For $i > 1$, the sequence reduces to,

$$\cdots \to H_i(S^{n-1}) \xrightarrow{i_\star} 0 \xrightarrow{\pi_\star} H_i(D^n, S^{n-1}) \xrightarrow{\partial_i} H_{i-1}(S^{n-1}) \xrightarrow{i_\star} 0 \to \cdots$$

Then by exactness, $\mathrm{im}(\pi_\star) = 0 = \ker(\partial_i)$ and $\ker(i_\star) = H_{i-1}(S^{n-1}) = \mathrm{im}(\partial_i) = \partial_i(H_i(D^n, S^{n-1}))$, so ∂_i is an isomorphism and for $i > 1$,

$$H_{i-1}(S^{n-1}) \cong H_i(D^n, S^{n-1})$$

Since the Δ-complex structure of S^{n-1} is one 0-cell attached to one $(n-1)$-cell we have, $H_{n-1}(S^{n-1}) = H_0(S^{n-1}) = \mathbb{Z}$ and $H_i(S^{n-1}) = 0$ otherwise. So for $i \neq 1$ we have,

$$H_i(D^n, S^{n-1}) \cong H_{i-1}(S^{n-1}) = \begin{cases} \mathbb{Z} & \text{i = n} \\ 0 & \text{otherwise} \end{cases}$$

And for $i = 1$ we have the following exact sequence,

$$\cdots \xrightarrow{i_\star} 0 \xrightarrow{\pi_\star} H_1(D^n, S^{n-1}) \xrightarrow{\partial_1} \mathbb{Z} \xrightarrow{i_\star} \mathbb{Z} \xrightarrow{\pi_\star} \cdots$$

Then we have, $\mathrm{im}(\partial_1) = \ker(i_\star) = 0$ and $\mathrm{im}(\pi_\star) = 0 = \ker(d_1)$. That is, $H_1(D^n, S^{n-1})$ must be the trivial group.

Hence, finally

$$H_i(D^n, S^{n-1}) = \begin{cases} \mathbb{Z} & \text{i = n} \\ 0 & \text{otherwise} \end{cases}$$

Problem 6. Show that if A is a retract of X then the map $H_n(A) \to H_n(X)$ induced by the inclusion $A \subset X$ is injective. Now show that there is no retraction from D^n to S^{n-1}, for any n.

Solution.

We know that chain maps between chain complexes induce homomorphisms between the homology groups of the respective complexes. If $A \subseteq X$ is a retract, then we have $i \circ r = \mathrm{id}_A \implies (i \circ r)_\star = (\mathrm{id}_A)_\star \implies i_\star$ is injective. So, $H_n(X) \to H_n(A) \to H_n(X)$ is injective.

Assume there is a retraction from D^n to S^{n-1}. Then $H_{n-1}(S^{n-1}) \xrightarrow{i_\star} H_{n-1}(D^n)$ is an injection. But $H_{n-1}(S^{n-1}) = \mathbb{Z}$ and $H_{n-1}(D^n) = 0$, a contradiction! Hence there is no retraction from D^n to S^{n-1} for any n.

Problem 7. Let $f : (X, A) \to (Y, B)$ be a map such that both $f : X \to Y$ and the restriction $f : A \to B$ are homotopy equivalences.

1. Show that
$$f_* : H_n(X, A) \to H_n(Y, B)$$
is an isomorphism for all n.

2. For the case of the inclusion
$$f : (D^n, S^{n-1}) \hookrightarrow (D^n, D^n - \{0\}),$$
show that f is not a homotopy equivalence of pairs — there is no
$$g : (D^n, D^n - \{0\}) \to (D^n, S^{n-1})$$
such that fg and gf are homotopic to the identity through maps of pairs. (Observe that a homotopy equivalence of pairs $(X, A) \to (Y, B)$ is also a homotopy equivalence for the pairs obtained by replacing A and B by their closures.)

Solution.

1. Since f is a homotopy equivalence we get the following commutative diagram (Corollary 2.11), where each sequence is long exact,
$$\begin{tikzcd}
\cdots \arrow[r] & H_i(A) \arrow[r, "i_*"] \arrow[d, "f_*"', "\cong"] & H_i(X) \arrow[r, "\pi_{1*}"] \arrow[d, "f_*"', "\cong"] & H_i(X, A) \arrow[r, "\partial_i"] \arrow[d, "f_*"] & H_{i-1}(A) \arrow[r] \arrow[d, "f_*"', "\cong"] & H_{i-1}(X) \arrow[r] \arrow[d, "f_*"', "\cong"] & \cdots \\
\cdots \arrow[r] & H_i(B) \arrow[r, "j_*"] & H_i(Y) \arrow[r, "\pi_{2*}"] & H_i(Y, B) \arrow[r, "\partial_i"] & H_{i-1}(B) \arrow[r] & H_{i-1}(Y) \arrow[r] & \cdots
\end{tikzcd}$$
Since the sequences are exact and we have the four isomorphisms induced by f, by the five lemma we know that f_* is also an isomorphism. Since i was arbitrary, this holds for all homologies and hence $f_* : H_n(X, A) \to H_n(Y, B)$ is an isomorphism for all n.

2. For the sake of contradiction assume f is a homotopy equivalence of pairs. Then there exists its homotopy inverse $g : (D^n, D^n - \{0\}) \to (D^n, S^{n-1})$ and we have
$$fg \simeq \mathrm{id}_Y \qquad gf \simeq \mathrm{id}_X$$
Since these homotopies are continuous maps and map $D^n - \{0\} \to D^n - \{0\}$ and $S^{n-1} \to S^{n-1}$ they must also map $\overline{D^n - \{0\}} \to \overline{D^n - \{0\}}$ and $\overline{S^{n-1}} \to \overline{S^{n-1}}$. Since

$\overline{D^n - \{0\}} = D^n$ and $\overline{S^{n-1}} = S^{n-1}$ we get another map of pairs,

$$f : (D^n, S^{n-1}) \to (D^n, D^n)$$

By part (1) we know that $H_i(D^n, S^{n-1}) \cong H_i(D^n, D^n)$ for all i. In particular, we know from Problem 5 that $H_n(D^n, S^{n-1}) = \mathbb{Z}$. But $H_n(D^n, D^n) = 0$, a contradiction!

Problem 8. Show that chain homotopy of chain maps is an equivalence relation.

Solution.

If P is a chain homotopy between the chain maps f and g, we know that,

$$\partial P + P\partial = g - f$$

Let P, Q be chain homotopies between f, g and g, h. Then,

1. **Reflexive:** Let $P = 0$. Then,

   $$\partial 0 + 0\partial = 0 = f - f$$

   Hence, $f \simeq_C f$.

2. **Symmetric:** Let $P^{-1} = -P$. Then we have,

   $$\partial(-P) + (-P)\partial = -(\partial P + P\partial) = -(g - f) = f - g$$

   Hence if $f \simeq_C g$, $g \simeq_C f$.

3. **Transitive:** To show that $f \simeq_C h$, we let $R = P + Q$. Then,

   $$\partial(P + Q) + (P + Q)\partial = g - f + h - g = h - f$$

   Hence if $f \simeq_C g$ and $g \simeq_C h$, we have $f \simeq_C h$.